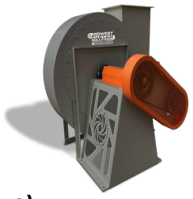
LETEK DCG / Bachoco - digestat

DEHYDRATION SYSTEM MASS-ENERGY BALANCE

	2.36	TPH wet chicken manure digestate cake (decanter c
	3.18	C.Y./hr @ 1485 lb/C.Y. and 80% moisture
	7x35	Single-pass rotary drum · XD-84 (7'×35')
Elevation	30	feet (above sea level)



1,576 Btu's/lb evaporation (rigorous)

Exhaust Gases	
11,242	ACFM
40	hp fan

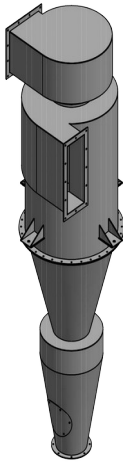
Dryer Outlet Gases	
11,242	ACFM
230	deg F

Fuel to Burner	
Propane	
19,900	Btu/lb
318	lb/hr
6.33	MMBtu/hr

Make-Up Air	
4,958	SCFM
85	deg F

Mixed Inlet Gases	
6,471	SCFM
900	deg F





Wet Feed	
80.0%	moisture
946	lb/hr solids
3,783	lb/hr water
4,729	lb/hr total

EVAPORATION RATE

3,616 LB/HR H2O

XD-84 (7'×35') · 35 ft long · 7.6 MMBtu/hr burner

Combustion Air	
1,456	ACFM
85	deg F
30%	X-S air

Dry Fuel (product burned)	
0	lb/hr total

Net Dry Product	
1,113	lb/hr total

Dry Material	
15.0%	moisture
946	lb/hr solids
167	lb/hr water
1,113	lb/hr total
180	deg F

MIDWEST CUSTOM ENGINEERING
PRECISION QUALITY INTEGRITY